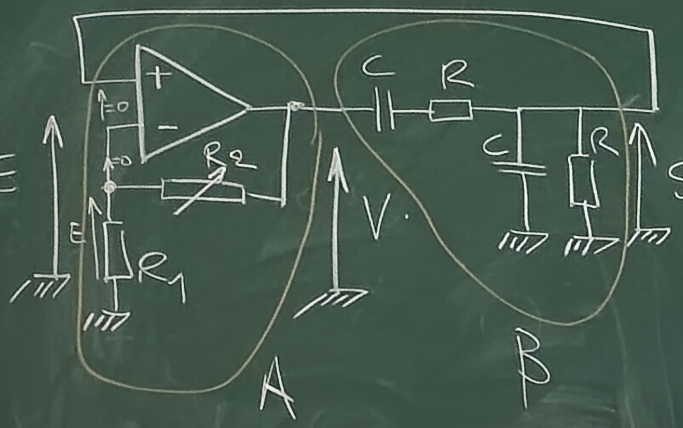
Oscillateur à pont de Wien



||H|| =

H(jω)

H =

On a :

E =

$\|H\| = 1$ et $\angle H = 0$: Condition de Barkhausen.

$\Downarrow (R_2 = 2R_1)$ $\rightarrow f_0$

ω_0 : pulsation d'oscillation. $z = a + jb$

$f_0 = \frac{\omega_0}{2\pi}$: fréquence d'oscillation $\|z\| =$

à la fréquence d'oscillation ($x = 1$)

$$\|H\| = \frac{A\,x}{\left[(1-x^2)^2 + (3x)^2\right]^{1/2}} = \frac{A}{3}$$

$\|H\| = 1 \Rightarrow \frac{A}{3} = 1 \Rightarrow 1 + \frac{R_2}{R_1} = 3 \Rightarrow \boxed{\frac{R_2}{R_1} = 2}$

$\frac{V}{E} = 1 +$

$S = V$

$z_2 = R$

$z_1 =$

RCw.

$(Cw)^2$

R_2

R_1

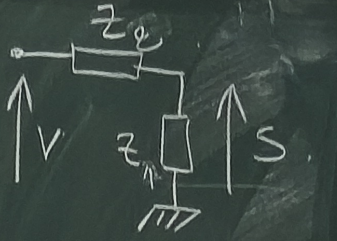
jx

réel ??

Barkhausen

$$\frac{V}{E} = 1 + \frac{R_2}{R_1} = A$$

$$S = V_0 \cdot \frac{z_1}{z_1 + z_2}$$



$$z_2 = R + \frac{1}{j c \omega}$$

$$z_1 = \frac{R \times \frac{1}{j c \omega}}{R + \frac{1}{j c \omega}} = \frac{R}{1 + j R C \omega}$$

V

$\beta =$

$=$

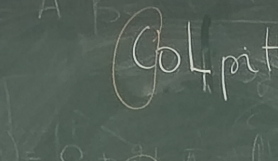
$=$

$$\beta = \frac{z_1}{z_1 + z_2} = \frac{\frac{R}{1+jRCw}}{\frac{R}{1+jRCw} + R + \frac{1}{jCw}} \times \frac{jCw(1+jRCw)}{jCw(1+jRCw)}$$

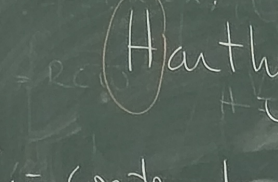
$$= \frac{jRCw}{jRCw + Rj^2Cw(1+jRCw) + 1+jRCw}$$

$$= \frac{jRCw}{1 + 3jRCw + (jRCw)^2}$$

Colpitts

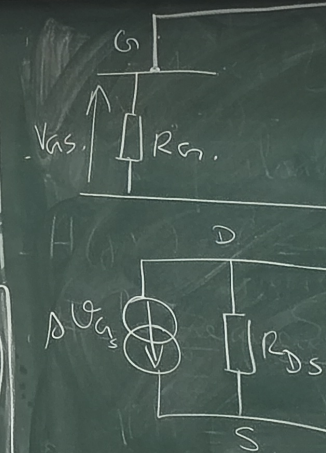


Hartly.

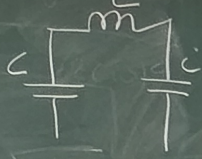


C_1 = condensateur de liaison

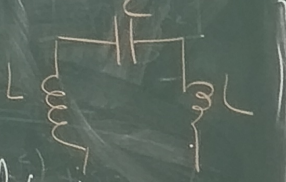
C_S = " de découplage.



Colpitts.

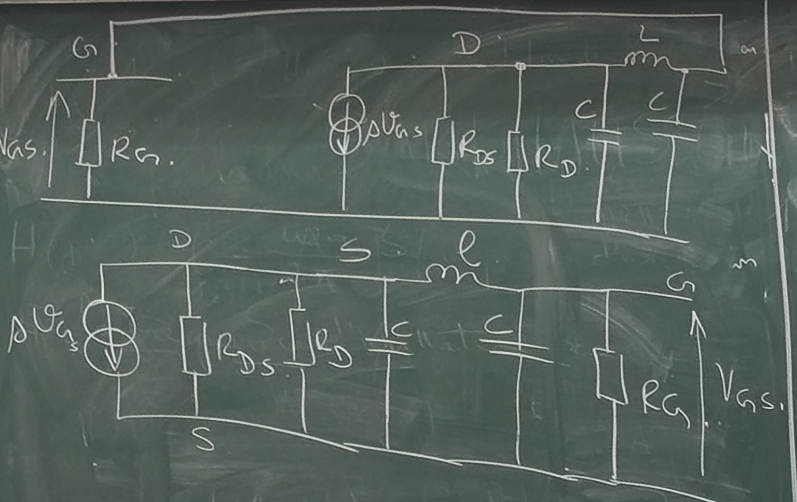


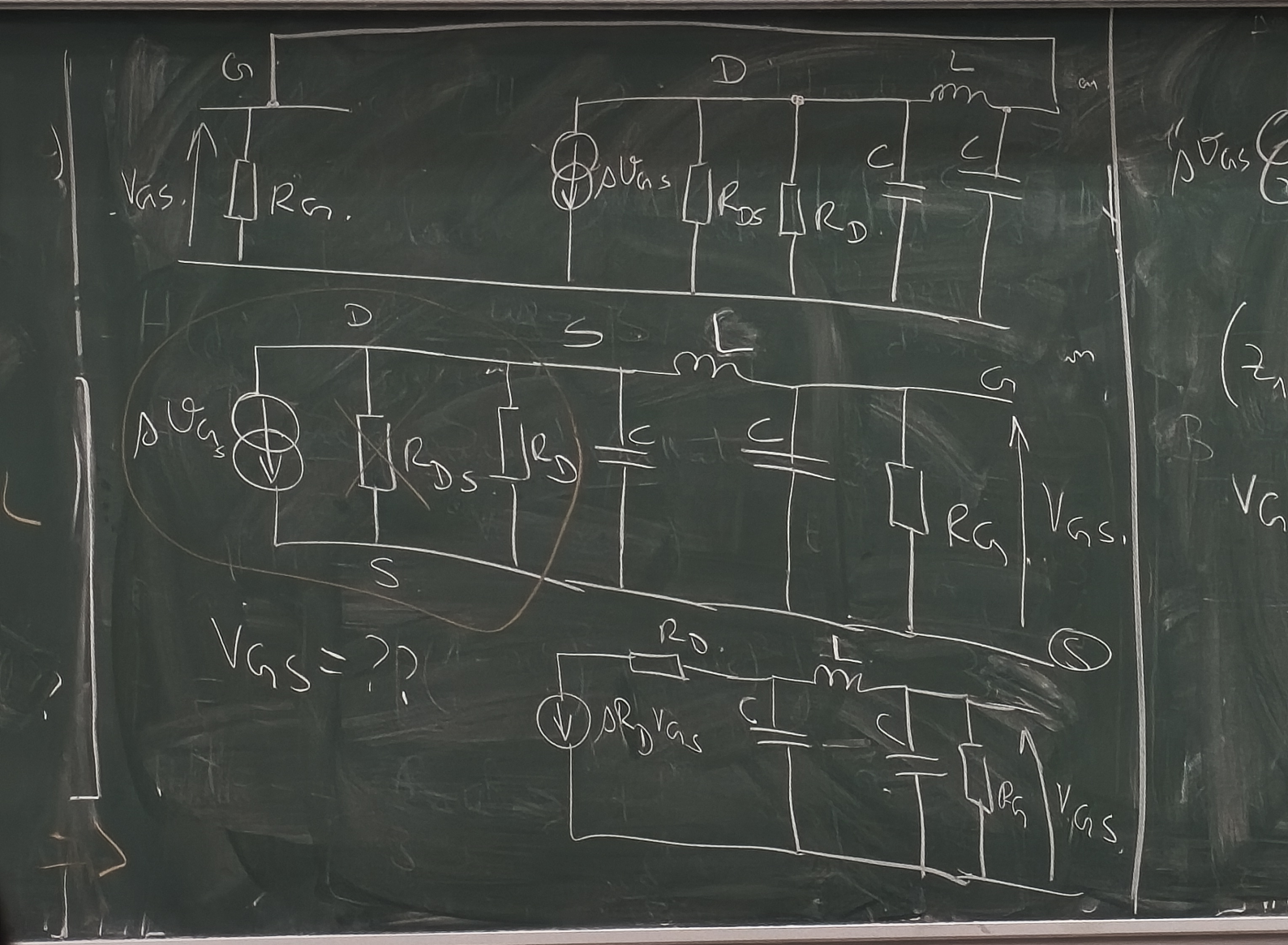
Hartley.



Montage de liaison

" de découplage.




G
D
L
V_{GS}
R_G
βV_{GS}
R_{DS}
R_D
C
C
D
S
L
G
βV_{GS}
R_{DS}
R_D
C
C
R_G
V_{GS}
S
$V_{GS} = ??$
R_D
L
$\beta R_D V_{GS}$
C
C
R_G
V_{GS}